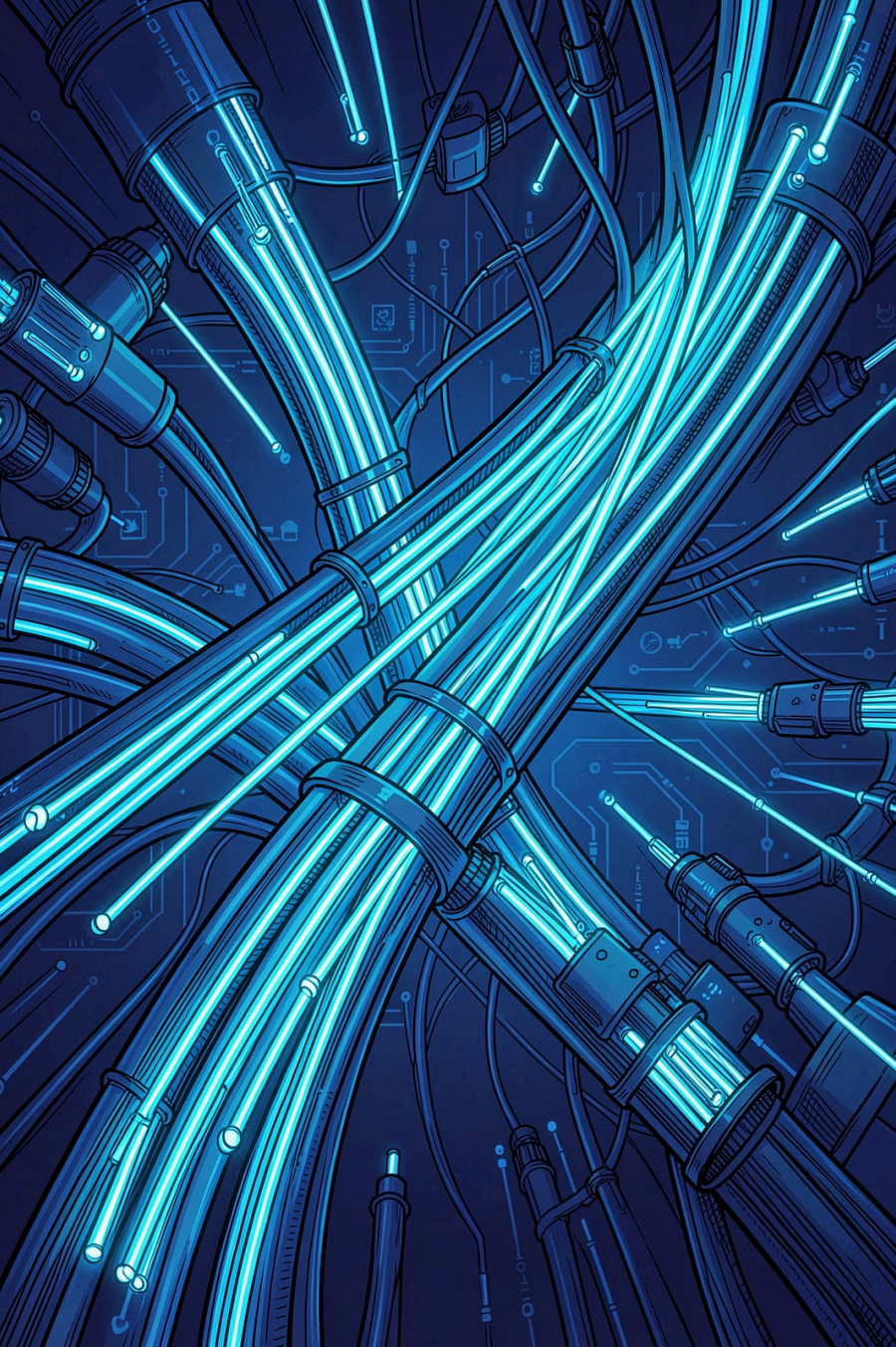




Telecom Customer Churn Prediction

End-to-End Data Analytics Pipeline · Hanieh Rezaei · June 2026

PYTHON

SQL SERVER

ML

POWER BI

Dataset at a Glance

6,418

Total Customers

28.8%

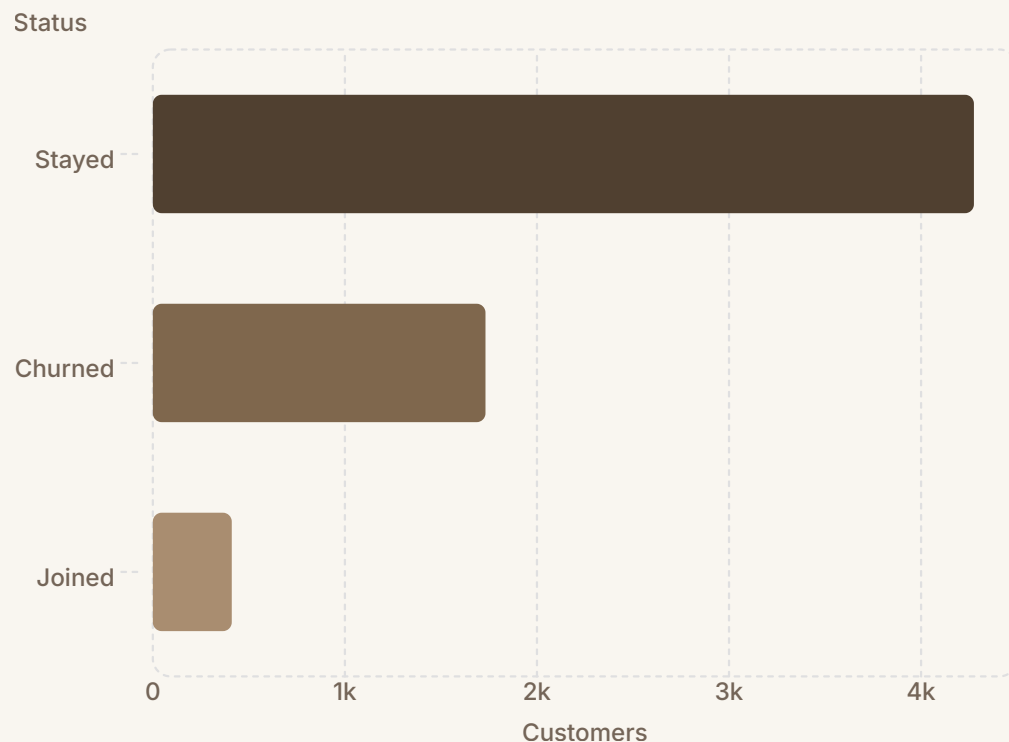
Churn Rate

411

New Joiners

32

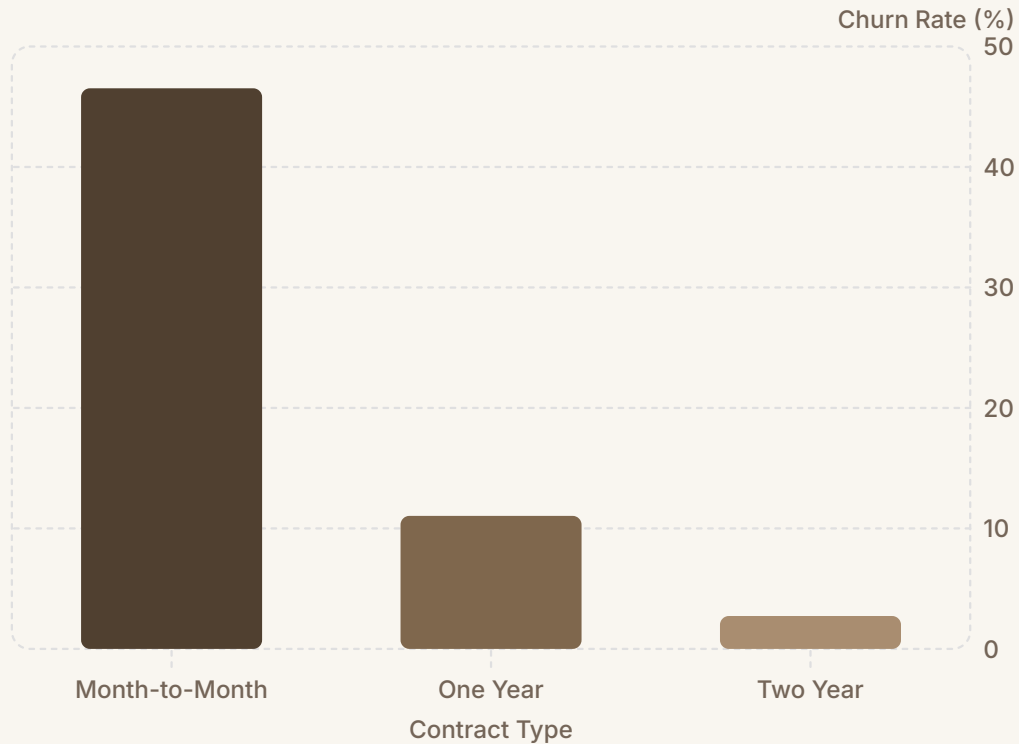
Features



Class Distribution

Moderately imbalanced — addressed via `class_weight='balanced'` (RF) and `scale_pos_weight` (XGBoost).

Key Churn Drivers



Contract Type = Strongest Predictor

Month-to-month customers churn at **46.5%** vs. **2.7%** for two-year contracts.

Other Key Findings

- Fiber Optic users: **41.1%** churn rate
- Older customers (>50): **31.6%** churn rate
- Jammu & Kashmir: **57.2%** churn rate
- Competition is the #1 churn reason

ML Pipeline & Model Performance

Feature Encoding

Apply encoders only on training folds then transform test.

Model Training

Train Random Forest and XGBoost models on folds.

Train/Test Split

Split data before any encoding to avoid leakage.

Evaluation & Tuning

5-fold GridSearchCV for metrics and hyperparameters.

Leakage-free design: split performed **before** label encoding.

Random Forest

Test AUC: **0.892** ·

Accuracy: **84%**

XGBoost

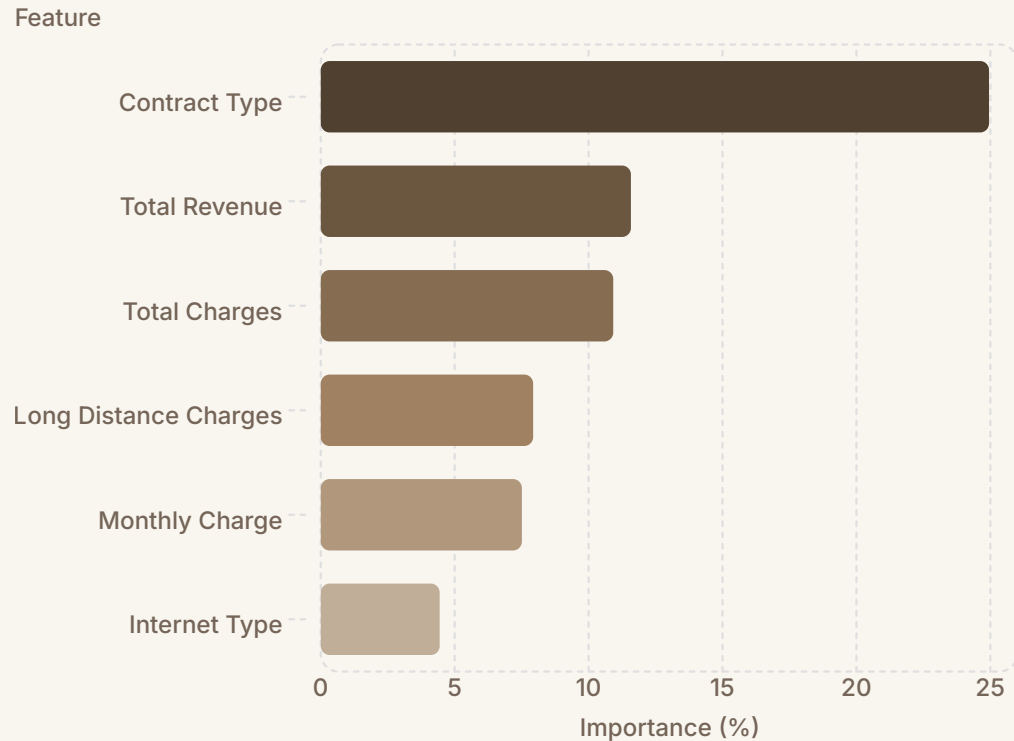
Test AUC: **0.897** ·

Accuracy: **82%**

Churner Recall: 77%

Most business-critical metric — missing a churner is costlier than a false alarm.

Feature Importance



Contract Type Dominates

Explains ~**25%** of all model decisions — confirming month-to-month contracts as the single strongest churn signal.

i Top 3 features account for **47.5%** of total importance.

New Customer Predictions

93.9%

At Risk of Churning

386 of 411 new customers

89%

Month-to-Month

vs. 49% in historical base

52%

MtM Churn Rate

Historical rate for this contract type

High predicted churn is **not a model error** — driven by structural risk: month-to-month contracts, no security, no premium support, lower charges.

Business Recommendations

0–3 Months: Act Now

- 1 Upgrade 386 at-risk customers to annual contracts; add free Online Security & Premium Support.

3–12 Months: Strategic

- 2 Outreach to high-probability churners; review Fiber Optic pricing; restructure onboarding.

Estimated Impact

- 3 Retain ~**247 customers**, protecting ~**\$750K** in lifetime revenue.

